Avatar Behavioral Authentication Dataset

Dataset Collection and Raw Data Documentation

Participants

The dataset consists of the temporal sequence observed for 16 volunteers (9 male, 7 female), aged 23–24 years, participated voluntarily with informed consent.

Collection Environment

Data were collected using a Meta Quest 2 headset in a custom Unity virtual art gallery. Participants remained seated and navigated using handheld controllers. The host system used an Intel Core i5 (9th Gen), 8 GB RAM, and NVIDIA GTX 1650 (4 GB). Tracking was recorded at 0.25 Hz (one sample every four seconds) with additional logging whenever the viewed image changed.

Interaction Protocol

Participants freely explored the gallery without any assigned task. Natural head movement and viewing behaviour were recorded. Individual sessions lasted approximately 2–10 minutes. Each participant contributed 2–11 sessions collected on the same day.

Dataset Statistics

Property	**Value**
Participants	16
Sessions	123
Total Duration	4 h 53 min
Average Session Length	2 min 23 sec
Sampling Rate	0.25 Hz
Raw Features	14

Raw Data Attributes

Feature	Description
UserID	Anonymous participant identifier
ImageName	Viewed image identifier
Order	Observation order
StartTime	Observation start time
EndTime	Observation end time
TotalTimeSpent	Elapsed session time
Direction	Movement direction
AngleChange	Head orientation change
HeadX	Head X coordinate
HeadY	Head Y coordinate
HeadZ	Head Z coordinate
DistanceFromStart	Distance from session start
DistanceFromPrevious	Distance from previous observation
Gender	Encoded gender

Notes

The released raw dataset is anonymized. Feature engineering, preprocessing, and model-ready datasets are documented separately.

Complete dataset is available at the following GitHub repository:

https://github.com/ravi-prakash1907/MetaPassAV-Passive-Behavior-Tracking-of-Avatars